Daily Sleep Prediction Using Wearable Sensors and Self-Reported Behavioral Metrics: A Multimodal Machine Learning Approach

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Introduction

- Robust sleep quality is essential for overall health and well-being [1].
- We identified behavioral and physiological features influencing sleep quality, including self-reported mood [2, 3], exercise [4], and phone usage [5].
- Leveraging consumer-accessible wearable devices enables our study to monitor sleep and related behaviors in natural, daily-life settings—enhancing generalizability to the public [8].
  - These devices allow scalable, non-invasive, and cost-effective analysis of sleep quality metrics.
- Our study examines combined feature interactions, which remain underrepresented in existing research [6–8], to identify trends in sleep and improve model accuracy for sleep optimization and future interventions.

Methods

- Dataset was drawn from TILES-2018 study, a 10-week collection of multimodal behavioral and physiological data from 212 hospital workers using Fitbit Charge 2 devices, smartphone EMAs, and other modalities [9].

Feature	Type	Units	Description
minutesAsleep	Sleep	Minutes	Total minutes spent asleep in a single sleep period.
efficiency	Sleep	%	% of time in bed spent asleep. Higher = better sleep utilization.
restfulness	Sleep	%	% of time in bed spent in REM + deep sleep. Higher = more restorative sleep.
Fat Burn_Minutes	Exercise	Minutes	Minutes per day spent in fat burn heart rate zone.
Cardio_Minutes	Exercise	Minutes	Minutes per day spent in cardio heart rate zone.
NumberSteps	Exercise	Steps	Total number of steps taken in a day.
dayPhoneDuration	Phone	Minutes	Total phone use time before sleep (resets daily).
lastSessionDuration	Phone	Minutes	Duration of the final phone session before sleep.
phoneSleepGap	Phone	Minutes	Minutes between end of last phone session and sleep start.
stress	Mood	EMA Score	Self-reported stress level (1 = lowest, 5 = highest).
anxiety	Mood	EMA Score	Self-reported anxiety level (1 = lowest, 5 = highest).

Figure 1. Features overview

- **Sleep features** were extracted as targets: *minutesAsleep, efficiency,* and *restfulness*; for simplicity, only the longest daily sleep session was retained

Three predictor groups were extracted: phone, exercise, and mood

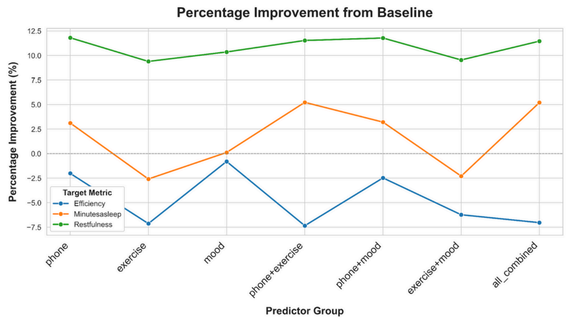
- **Phone features**—*dayPhoneDuration, phoneSleepGap, and lastSessionDuration*— were based on RealizD data, an app that passively tracks phone data. Phone sessions < 30 seconds were removed for clarity
- **Exercise feature** included *Fat Burn_minutes*, *Cardio_minutes*, and *NumberSteps* from Fitbit data
- **Mood features**—self-reported *stress* and *anxiety* (1-5 scale)—were gathered using EMAs.

- Records with minutesAsleep = 0 or missing phone data were removed; all other missing values were median-imputed.
- We trained Random Forest (RF) models using 10-fold cross-validation grouped by participant ID and compared them to a baseline mean-prediction model using RMSE [12].
- Each model used a different combination of predictor groups and was evaluated on *restfulness*, *minutesAsleep*, and *efficiency*.
- The seven predictor groupings tested:
  - 1—phone, 2—exercise, 3—mood, 4—phone + exercise, 5—phone + mood, 6—exercise + mood, 7—all features combined
- RF hyperparameters: 100 trees, max depth = 10, min samples per leaf = 5, random seed = 42.
- This design enabled systematic comparison of how behavioral and physiological factors influence different sleep metrics [13].

Results

Figure 2.
Results overview

Percent decrease in RMSE from baseline (positive values indicate better performance) for Random Forest models across the three sleep metrics.



Restfulness

- Most predictable metric across all predictor groups.
- Best RF result: phone features only, 11.8% improvement over baseline.
- Notable combined group results: phone + mood: 11.78%, phone + exercise: 11.6%, exercise + mood: 9.5%.
- Consistent gains suggest restfulness is strongly tied to daily behavioral signals (i.e. phone use, self-reported mood) [10].

Minutes Asleep

- More modest and variable improvements.
- Best RF models: phone + exercise and all features (~+5.1% over baseline).
- Other results: phone only: +3.2%, phone + mood: +3.3%, exercise-only and exercise + mood: ~−2.4% (below baseline)
- Indicates daily behaviors aid prediction, but individual variability and environmental noise reduce accuracy [11].

Efficiency

- Most difficult metric to model.
- All RF models underperformed baseline (–1% to −7.4% RMSE).
- Worst results: phone + exercise; least decline: mood-only.
- Likely influenced by external, unmeasured factors (i.e. room temperature, sleep disorders) [12].

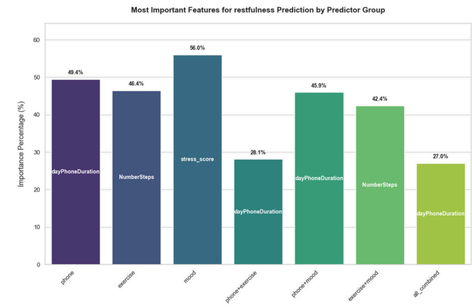


Figure 3.
Feature Importance for Random Forest Models predicting *restfulness* across feature groups

dayPhoneDuration was the most predictive feature across most groups, reaching 49.4% importance in the phone group and remaining above 27% in all models.
NumberSteps (46.4%) and *stress* (56%) were the strongest predictors within the exercise and mood groups, respectively.

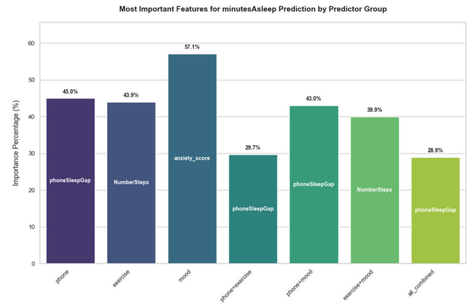


Figure 4.
Feature Importance for Random Forest Models predicting *minutesAsleep* across feature groups

phoneSleepGap was the most predictive feature across most groups, reaching 45% importance in the phone group. Similar to restfulness,
NumberSteps (44%) was the strongest predictor within the exercise groups.
Within the mood group, *anxiety* (57%) was more predictive than *stress*.

Conclusions

- Models trained on multimodal wearable and self-reported data can improve prediction of *restfulness* and, to a lesser extent, *minutesAsleep.*
  - Highlights potential of models to identify key features for sleep prediction.
- Supports practical, user-oriented approaches.
  - Daily-collected, wearable and self-report data; use of implementable models.
- Highlights need for additional variables, alternative models, and longer-term monitoring to improve prediction of complex sleep outcomes.

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